Link Layer

In the link layer, there are two different types of link-layer communication channels. The first type are **broadcast channels**, these channels connect multiple hosts in wireless and wired networks such as LANs, satellite networks, and hybrid fiber-coaxial cable (HFC) access networks. Since many hosts are connected to the same broadcast communication channel, a so-called ***medium access protocol*** is needed to coordinate data transmission. In some cases, a central controller device may be used to coordinate transmissions; in other cases, the hosts themselves coordinate transmissions. The second type of link-layer channel is the **point-to-point communication link**, such as that often found between two routers connected by a long-distance link, or between a user's office computer and the nearby Ethernet switch to which it is connected. Coordinating access to a point-to-point link is simpler and is done using the Point-to-Point Protocol (PPP). PPP is used in settings ranging from dial-up service using a telephone line to high-speed point-to-point data transport over fiber-optic links.

Before moving on. Let us discuss some important terminology:

- Node: any device that runs a link-layer (i.e., layer 2) protocol. Nodes include hosts, routers, switches, and WiFi access points
- Link: the communication channels that connect adjacent nodes along the communication path.

In order for an IP layer datagram to be transferred from source host to destination host, it must be moved over each of the individual links in the end-to-end path. Over a given link, a transmitting node encapsulates the datagram into **a link-layer frame** and transmits the frame into the communication link.

Example: In order to gain further insight into the link layer and how it relates to the network layer, let's consider a transportation analogy. Consider a travel agent who is planning a trip for a tourist traveling from Lahore, Pakistan, to Lausanne, Switzerland. The travel agent decides that it is most convenient for the tourist to take a Suzuki Cultus from Lahore to Islamabad airport, then a plane from Islamabad airport to Geneva's airport, and finally a train from Geneva's airport to Lausanne's train station. Once the travel agent makes the three reservations, it is the responsibility of the Travel agent's company to get the tourist from Lahore to Islamabad; it is the responsibility of the airline company to get the tourist from Islamabad airport to Geneva; and it is the responsibility of the Swiss train service to get the tourist from Geneva to Lausanne. Each of the three segments of the trip is "direct" between two "adjacent" locations. Note that the three transportation segments are managed by different companies and use entirely different transportation modes (limousine, plane, and train). Although the transportation modes are different, they each provide the basic service of moving passengers from one location to an adjacent location. **In this transportation analogy, the tourist is a datagram, each transportation segment is a link, the transportation mode is a link layer protocol, and the travel agent is a routing protocol.**

Services offered by Link Layer:

Although the basic service of any link layer is to move a datagram from one node to an adjacent node over a single communication link, the details of the provided service can vary from one link-layer protocol to the next. Possible services that can be offered by a link-layer protocol include:

- Framing. Almost all link-layer protocols encapsulate each network-layer datagram within a link-layer frame before transmission over the link. A frame consists of a data field, in which the network-layer datagram is inserted, and a number of header fields. The structure of the frame is specified by the link-layer protocol.
- Link access. A medium access control (MAC) protocol specifies the rules by which a frame is transmitted onto the link. For point-to-point links that have a single sender at one end of the link and a single receiver at the other end of the link, the MAC protocol is simple (or nonexistent)—the sender can send a frame whenever the link is idle. The more interesting case is when multiple nodes share a single broadcast link—the so-called multiple access problem. Here, the MAC protocol serves to coordinate the frame transmissions of the many nodes.

  Question: What is the work of MAC protocol?
- Reliable delivery. When a link-layer protocol provides reliable delivery service, it guarantees to move each network-layer datagram across the link without error. Recall that certain transport-layer protocols (such as TCP) also provide a reliable delivery service. Similar to a transport-layer reliable delivery service, a link-layer reliable delivery service can be achieved with acknowledgments and retransmissions (see Section 3.4). A link-layer reliable delivery service is often used for links that are prone to high error rates, such as a wireless link, with the goal of correcting an error locally—on the link where the error occurs—rather than forcing an end-to-end retransmission of the data by a transport- or application-layer protocol. However, link-layer reliable delivery can be considered an unnecessary overhead for low bit-error links, including fiber, coax, and many twisted-pair copper links. For this reason, many wired link-layer protocols do not provide a reliable delivery service.
- Error detection and correction. The link-layer hardware in a receiving node can incorrectly decide that a bit in a frame is zero when it was transmitted as a one, and vice versa. Such bit errors are introduced by signal attenuation and electromagnetic noise. Because there is no need to forward a datagram that has an error, many link-layer protocols provide a mechanism to detect such bit errors. This is done by having the transmitting node include error-detection bits in the frame, and having the receiving node perform an error check.

Recall that the Internet's transport layer and network layer also provide a limited form of error detection—the Internet checksum. Error detection in the link layer is usually more sophisticated and is implemented in hardware. Error correction is similar to error detection, except that a receiver not only detects when bit errors have occurred in the frame but also determines exactly where in the frame the errors have occurred (and then corrects these errors).

Where is the Link Layer implemented?

The Ethernet capabilities are either integrated into the motherboard chipset or implemented via a low-cost dedicated Ethernet chip. For the most part, the link layer is implemented on a chip called the network adapter, also sometimes known as a network interface controller (NIC). The network adapter implements many link-layer services including framing, link access, error detection, and so on. Thus, much of a link-layer controller's functionality is implemented in hardware.

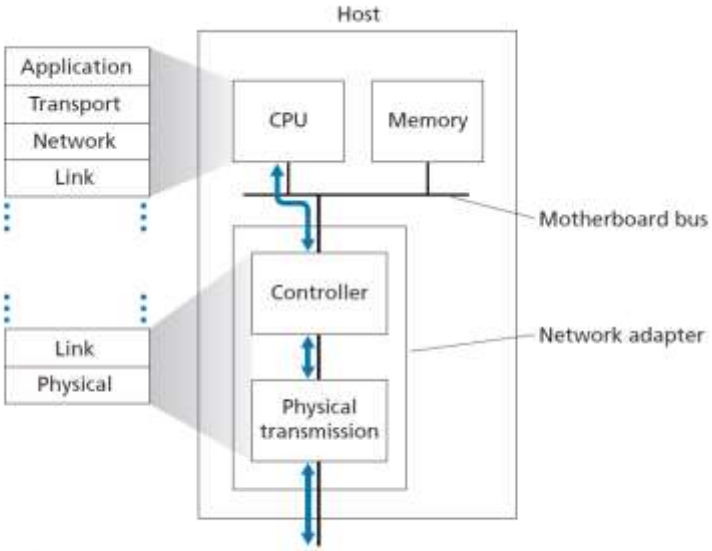


Figure 6.2 ♦ Network adapter: Its relationship to other host components and to protocol stack functionality

On the sending side, the controller takes a datagram that has been created and stored in host memory by the higher layers of the protocol stack, encapsulates the datagram in a link-layer frame (filling in the frame's various fields), and then transmits the frame into the communication link, following the link-access protocol. On the receiving side, a controller receives the entire frame, and extracts the network layer datagram. If the link layer performs error detection, then it is the sending controller that sets the error-detection bits in the frame header and it is the receiving controller that performs error detection. Figure 6.2 shows that while most of the link layer is implemented in hardware, part of the link layer is implemented in software that runs on the host's CPU. The software components of the link layer implement higher-level link-layer functionality such as assembling link-layer addressing information and activating the controller hardware. On the receiving side, link-layer software responds to controller interrupts (for example, due to the receipt of one or more frames), handling error conditions and passing a datagram up to the network layer. Thus, the link layer is a combination of hardware and software—the place in the protocol stack where software meets hardware.

Error Detection and Correction

As we already see, bit-level error detection and correction - detecting and correcting the corruption of bits in a link-layer frame sent from one node to another physically connected neighboring node—are two services often provided by the link layer. We saw that error-detection and

correction services are also often offered at the transport layer as well. Next, we'll see some techniques of error detection and correction at link layer level.

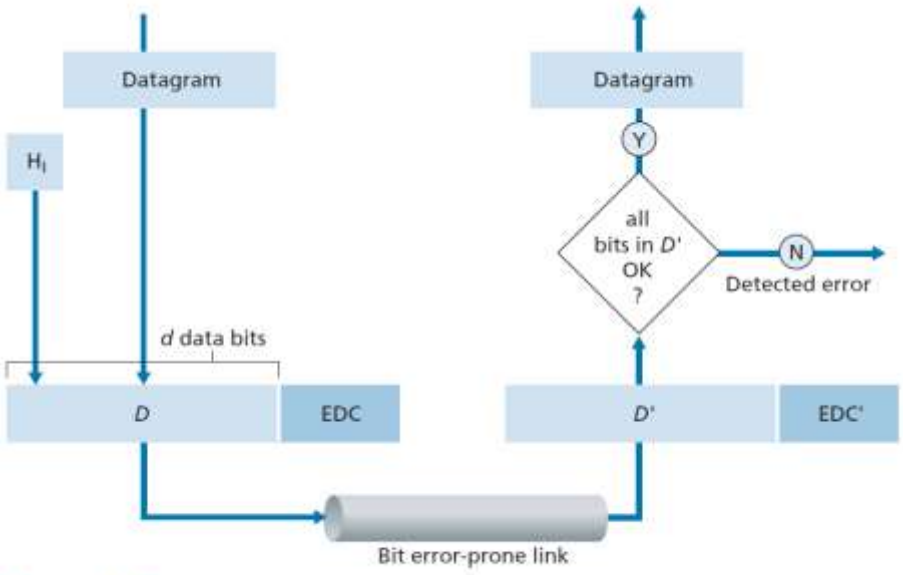


Figure 6.3 ♦ Error-detection and -correction scenario

Figure 6.3 illustrates the setting for our study. At the sending node, data, D, to be protected against bit errors is augmented with error-detection and correction bits (EDC). Typically, the data to be protected includes not only the datagram passed down from the network layer for transmission across the link, but also link-level addressing information, sequence numbers, and other fields in the link frame header. Both D and EDC are sent to the receiving node in a link-level frame. At the receiving node, a sequence of bits, D' and EDC' is received. Note that D' and EDC' May differ from the original D and EDC as a result of in-transit bit flips. The receiver's challenge is to determine whether or not D! is the same as the original D, given that it has only received D! and EDC'. The exact wording of the receiver's decision in Figure 6.3 (we ask whether an error is detected, not whether an error has occurred!) is important. Error-detection and correction techniques allow the receiver to sometimes, but not always, detect that bit errors have occurred. Even with the use of error-detection bits there still may be undetected bit errors; that is, the receiver may be unaware that the received information contains bit errors. As a consequence, the receiver might deliver a corrupted datagram to the network layer, or be unaware that the contents of a field in the frame's header has been corrupted. We thus want to choose an error-detection scheme that keeps the probability of such occurrences small. Generally, more sophisticated error-detection and correction techniques (that is, those that have a smaller probability of allowing undetected bit errors) incur a larger overhead—more computation is needed to compute and transmit a larger number of error-detection and correction bits.

What is the difference between Router and Switch?

Routers are store-and-forward packet switches that forward packets using network-layer addresses. Although a switch is also a store-and forward packet switch, it is fundamentally different from a router in that it forwards packets using MAC addresses. Whereas a router is a layer-3 packet switch, a switch is a layer-2 packet switch.

Link Layer Addressing:

Hosts and routers have link-layer addresses. Now you might find this surprising, recalling from Chapter 4 that hosts and routers have network-layer addresses as well.

You might be asking, why in the world do we need to have addresses at both the network and link layers?
Before diving into this question. Let us first discuss what is a MAC address.

Medium Access Control (MAC) Address

In truth, it is not hosts and routers that have link-layer addresses but rather their adapters (that is, network interfaces) that have link-layer addresses. A host or router with multiple network interfaces will thus have multiple link-layer addresses associated with it, just as it would also have multiple IP addresses associated with it. It's important to note, however, that link-layer switches do not have link-layer addresses associated with their interfaces that connect to hosts and routers. This is because the job of the link-layer switch is to carry datagrams between hosts and routers; a switch does this job transparently, that is, without the host or router having to explicitly address the frame to the intervening switch. This is illustrated in Figure 6.16. A link layer address is also called a LAN address, a physical address, or a MAC address. For most LANs (including Ethernet and 802.11 wireless LANs), the MAC address is 6 bytes long, giving 248 possible MAC addresses.

An adapter's MAC address has a flat structure (as opposed to a hierarchical Structure of IP) and doesn't change no matter where the adapter goes. A laptop with an Ethernet interface always has the same MAC address, no matter where the computer goes. A smartphone with an 802.11 interface always has the same MAC address, no matter where the smartphone goes.

An Example of difference between MAC address and IP

An adapter's MAC address is analogous to a person's CNIC number, which also has a flat addressing structure and which doesn't change no matter where the person goes. An IP address is analogous to a person's postal address, which is hierarchical and which must be changed whenever a person moves. Just as a person may find it useful to have both a postal address and a social security number, it is useful for a host and router interfaces to have both a network-layer address and a MAC address.

When an adapter wants to send a frame to some destination adapter, the sending adapter inserts the destination adapter's MAC address into the frame and then sends the frame into the LAN. Thus, when an adapter receives a frame, it will check to see whether the destination MAC address in the frame matches its own MAC address. If there is a match, the adapter extracts the enclosed datagram and passes the datagram up the protocol stack. If there isn't a match, the adapter discards

the frame, without passing the network-layer datagram up. Thus, the destination only will be interrupted when the frame is received.

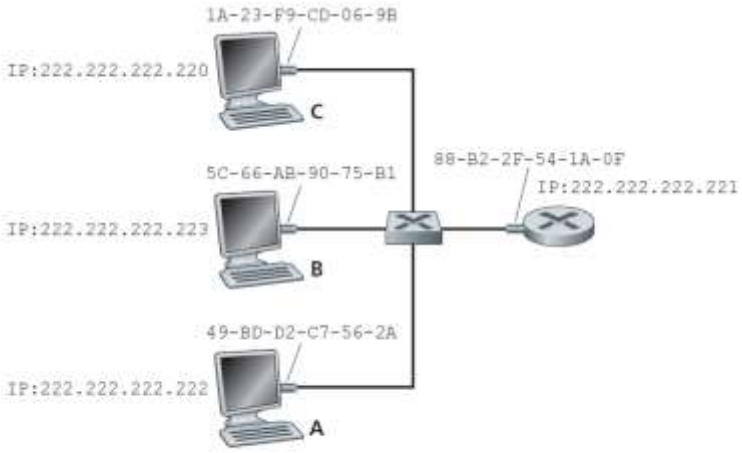


Figure 6.17 • Each interface on a LAN has an IP address and a MAC address

What is the work of Address Resolution Protocol (ARP)?

Because there are both network-layer addresses (for example, Internet IP addresses) and link-layer addresses (that is, MAC addresses), there is a need to translate between them. For the Internet, this is the job of the Address Resolution Protocol (ARP).

To understand the need for a protocol such as ARP, consider the network shown in Figure 6.17. In this simple example, each host and router has a single IP address and single MAC address. As usual, IP addresses are shown in dotted-decimal notation and MAC addresses are shown in hexadecimal notation. For the purposes of this discussion, we will assume in this section that the switch broadcasts all frames; that is, whenever a switch receives a frame on one interface, it forwards the frame on all of its other interfaces.

Now suppose that the host with IP address 222.222.222.220 wants to send an IP datagram to host 222.222.222.222. In this example, both the source and destination are in the same network. To send a datagram, the source must give its adapter not only the IP datagram but also the MAC address for destination 222.222.222.222. The sending adapter will then construct a link-layer frame containing the destination's MAC address and send the frame into the LAN.

The important question addressed in this section is, How does the sending host determine the MAC address for the destination host with IP address 222.222.222.222?

As you might have guessed, it uses ARP. An ARP module in the sending host takes any IP address on the same LAN as input, and returns the corresponding MAC address. In the example at hand, sending host 222.222.222.220 provides its ARP module the IP address 222.222.222.222, and the ARP module returns the corresponding MAC address 49-BD-D2-C7-56-2A. So we see that ARP resolves an IP address to a MAC address. In many ways it is analogous to DNS (studied in Section

2.5), which resolves host names to IP addresses. However, one important difference between the two resolvers is that DNS resolves host names for hosts anywhere in the Internet, whereas ARP resolves IP addresses only for hosts and router interfaces on the same network. If a node in Lahore were to try to use ARP to resolve the IP address for a node in Karachi, ARP would return with an error.

Link Layer Switches

The role of the switch is to receive incoming link-layer frames and forward them onto outgoing links. We'll see that the switch itself is **transparent** to the hosts and routers in the network; that is, a host/router addresses a frame to another host/router (rather than addressing the frame to the switch) and happily sends the frame into the LAN (Local Area Network), unaware that a switch will be receiving the frame and forwarding it. The rate at which frames arrive to any one of the switch's output interfaces may temporarily exceed the link capacity of that interface. To accommodate this problem, switch output interfaces have buffers, in much the same way that router output interfaces have buffers for datagrams. Filtering and forwarding are the two basic functions that are performed by the switch.

Filtering is the switch function that determines whether a frame should be forwarded to some interface or should just be dropped. **Forwarding** is the switch function that determines the interfaces to which a frame should be directed, and then moves the frame to those interfaces. Switch filtering and forwarding are done with a **switch table**. The switch table contains entries for some, but not necessarily all of the hosts and routers on a LAN. An entry in the switch table contains (1) a MAC address, (2) the switch interface that leads toward that MAC address, and (3) the time at which the entry was placed in the table. Note that switches forward packets based on MAC addresses rather than on IP addresses.